

CN ASSIGNMENT - II

Name $\Rightarrow$ Janmejai Prastap Tonk

University $\Rightarrow$ 2025243
Roll no -

Roll no $\Rightarrow$ 31

Section $\Rightarrow$ C

Semester $\Rightarrow$ 5th

Ans. | UDP Socket System Calls: UDP is connectionless, so the client and server do not establish a connection before communication.

UDP Server Sequence:

Socket() → bind() → recvfrom() → Sendto() → Close()

- ① Socket() → Creates a UDP Socket
- ② bind() → Associates the socket with a specific IP address and port no.
- ③ recvfrom() → Waits for and receives a datagram from a client.
- ④ Sendto() → Sends a datagram back to the client.
- ⑤ Close() → Closes the socket.

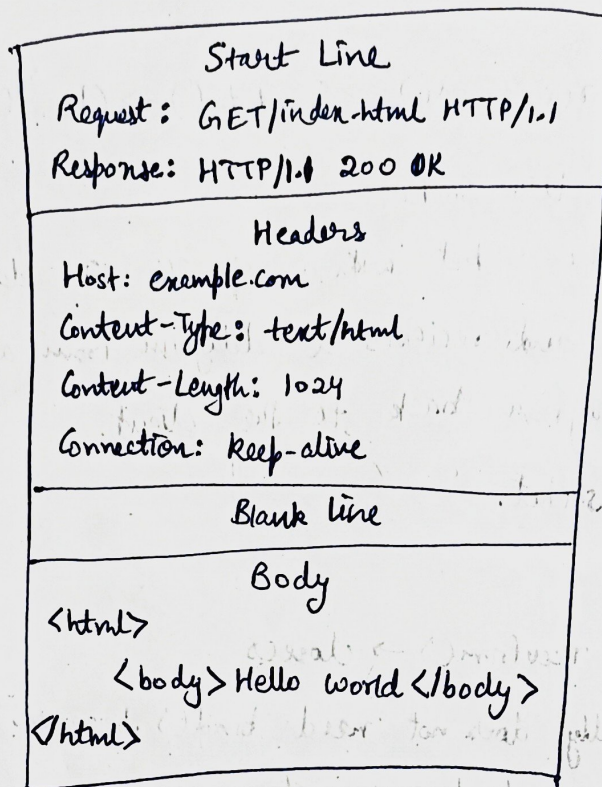
UDP Client Sequence:

Socket() → sendto() → recvfrom() → close()

The UDP client normally does not need bind(), because the operating system can automatically assign a temporary port.

UDP	TCP
① Connectionless Protocol.	① Connection-oriented protocol.
② Server uses socket() → bind() → recvfrom() / sendto().	② Server uses socket() → bind() → listen() → accept().
③ Client uses socket() → sendto() / recvfrom().	③ Client uses socket() → connect() → send() / recv().
④ No connection establishment.	④ Requires connection establishment.
⑤ Uses sendto() and recvfrom().	⑤ Uses send() and recv() after connection.

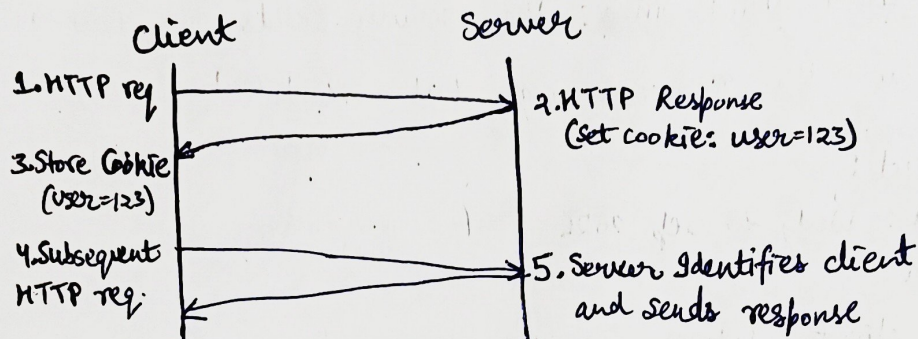
A.2 An HTTP message is used for communication between a client and a server. It consists of a start line, headers, a blank line, and an optional message body.



Components :

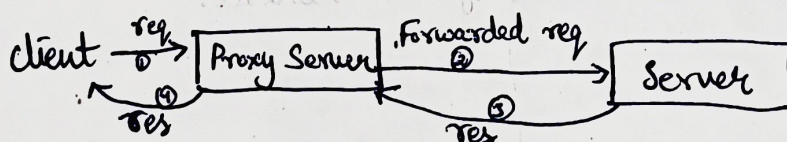
- ① **Start Line:** The first line of an HTTP message. In a request, it contains the HTTP method, resource, and version, such as GET/index.html HTTP/1.1. In a response, it contains the HTTP version, status code, and status description, such as HTTP/1.1 200 OK.
- ② **Headers:** Headers contain additional information about the request or response, such as host, content type, content length and connection details.
- ③ **Blank Line:** A blank line separates the headers from the message body. It indicates that the header section has ended.
- ④ **Body:** The Body contains the actual data being transferred. It may contain HTML, JSON, images, text, or other content. The body may be absent in some HTTP messages.

Ans-3 A cookie is a small piece of data sent by a server and stored by the client's browser. It helps the server recognize the client when it receives subsequent requests.



So cookies allow HTTP, which is otherwise stateless, to maintain information across multiple requests.

Ans-4 A proxy server is an intermediary server that acts between a client and the destination server. Instead of communicating directly with the destination server, the client sends its request to the proxy, which forwards the request and returns the server's response to the client.



Functions of a Proxy Server:

- ① Intermediary: Acts as a middle layer between client and server.
- ② Caching: Can store frequently requested resources and serve them without contacting the destination server again.
- ③ Filtering: Can control or filter client requests.
- ④ Security: Can provide an additional layer between clients and external servers.
- ⑤ Access Controls: Can restrict access to certain websites or resources.

Ans-5 HTTP cache validation allows a client to check whether a cached resource has changed on the server before downloading it again.

- ① Last-Modified / If-Modified-Since: The server sends the Last-Modified header indicating when the resource was last modified.

Server → Client:

Last-Modified: Wed, 23 Sep 2026
10:00:00 GMT

When the client later requests the same resource, it sends:

If-Modified-Since: Wed, 23 Sep 2026
10:00:00 GMT

If the resource has not changed, the server responds:

304 Not Modified

The client can then use its cached copy.

- ② ETag / If-None-Match: The server assigns a unique identifier, called an ETag to a particular version of the resource.

Server → Client:

ETag: "abc123"

For a later request, the client sends:

If-None-Match: "abc123"

If the resource still has the same ETag, the server responds:

304 Not Modified

The client can use its cached copy.

Reliable: ETag / If-None-Match is generally more reliable, for identifying whether the representation has changed because it compares an identifier for the specific resource version rather than relying only on modification time.

Ans-6 ① SMTP - Simple Mail Transfer Protocol:

- It is ~~not~~ mainly used for sending and transferring emails.
- * It is used:
- To send an email from a client to a mail server.
 - To transfer emails between mail servers.

② POP3 - Post Office Protocol Version 3:

POP3 is used for retrieving emails from a mail server to a user's email client.

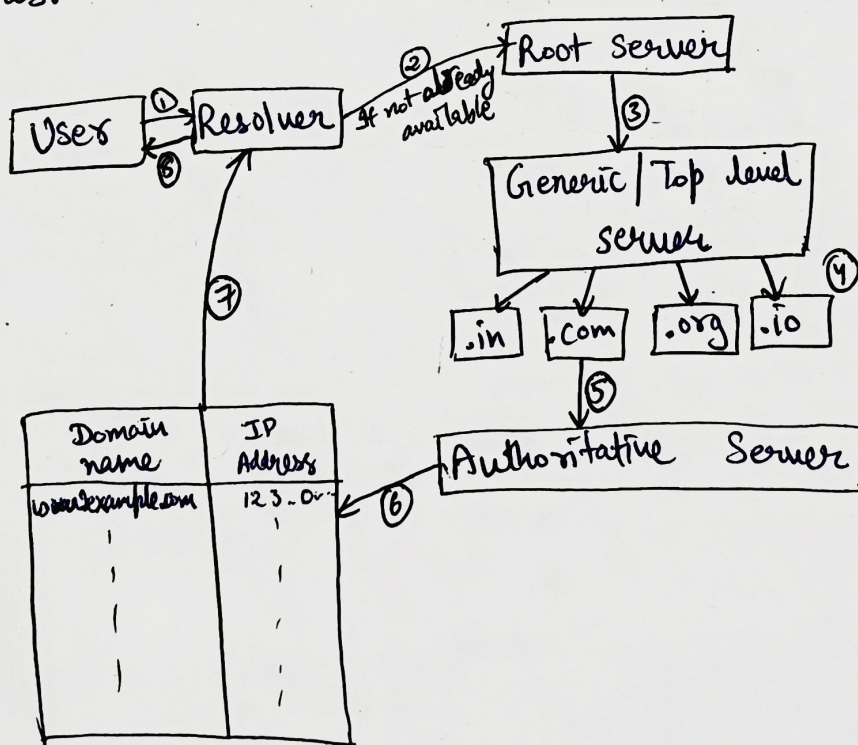
It generally downloads messages to the client, and depending on the configuration, messages may be removed from the server.

③ IMAP - Internet Message Access Protocol:

IMAP is also used for retrieving and managing emails, but it keeps the messages on the mail server and allows the client to synchronize ~~them~~ them.

This makes it suitable when the same mailbox is accessed from multiple devices.

Ans-7



Aug-8 DNS records store different types of information about a domain name.

Record	Full Form	Purpose
A	Address Record	Maps a domain name to an IPv4 address.
AAAA	IPv6 Address Record	Maps a domain name to an IPv6 address.
CNAME	Canonical Name	Creates an alias from one domain name to another domain name.
MX	Mail Exchange	Specifies the mail servers responsible for receiving email for a domain.
NS	Name Servers	Specifies the authoritative server DNS servers for a domain.
PTR	Pointer	Maps an IP address back to a domain name.
Txt	Text	Stores arbitrary text information, commonly used for domain verification and email security policies.
SOA	Start of Authority	Contains important administrative information about a DNS zone, such as the primary name server and serial number.

Ans-9 Optimistic unchoking is a mechanism used in BitTorrent where a peer temporarily uploads data to a randomly selected peer, even if that peer is not currently uploading data back.

In the normal tit-for-tat strategy, a peer prefers to upload to other peers that provide good upload rates in return. However, a peer periodically selects another peer optimistically and allows it to download data temporarily.

Necessary:

- ① Gives new peers a ~~chance~~ chance.
- ② Discovers better peers.
- ③ Prevent permanent exclusion.
- ④ Improves overall performance.

Ans-10

Given, $F = 20 \text{ Mb}$, $u_s = 400 \text{ Kbps}$, $d_{\min} = 200 \text{ Kbps}$, $D_{cs} = ?$, $N = 5, 10, 20, 40, 80$.

$$\text{So, } \frac{F}{d_{\min}} = \frac{20000 \text{ Kb}}{200 \text{ Kbps}} = 100 \text{ s}$$

$$\frac{NF}{u_s} = \frac{N(20000)}{400} = 50 \text{ Ns}$$

$$\text{So, } D_{cs} = \max\left(\frac{NF}{u_s}, \frac{F}{d_{\min}}\right)$$

$$D_{cs} = \max(50N, 100)$$

Table:

N	NF/u _s	F/d _{min}	D _{cs}
5	250 s	100 s	250 s
10	500 s	100 s	500 s
20	1000 s	100 s	1000 s
40	2000 s	100 s	2000 s
80	4000 s	100 s	4000 s

The distribution time increases linearly with N because the server must upload the entire file to every client. As the number of clients increases, the server's upload capacity becomes the bottleneck.

Ans-11

$$D_{\text{par}} = \max \left(\frac{F}{u_s}, \frac{F}{d_{\min}}, \frac{NF}{u_s + \sum_{i=1}^N u_i} \right)$$

Given, $F = 20 \text{ Mb} = 2000 \text{ Kb}$, $u_s = 400 \text{ Kbps}$, $d_{\min} = 200 \text{ Kbps}$

$$D_{\text{par}} = \max \left(50 \text{ s}, 100 \text{ s}, \frac{20000N}{400 + 100N} \right)$$

Table:

N	F/u _s	F/d _{min}	NF/u _s +N u _i	D _{par}
5	50 s	100 s	200 s	200 s
10	50 s	100 s	285.71 s	285.71 s
20	50 s	100 s	363.64 s	363.64 s
40	50 s	100 s	390.24 s	390.24 s
80	50 s	100 s	395.06 s	395.06 s

In client-server architecture, only the server contributes upload bandwidth, therefore:

$$D_{cs} \propto N$$

As the no. of client increases, the server has to upload more copies.

In P2P, every peer can also upload pieces of the file.

- Therefore, the total available upload capacity becomes:

$$U_s + N u_i$$

As N increases, more peers contribute upload bandwidth.

Ans. 12 (a) Non Persistent

① New TCP connection for each object.

② More connection-establishment overhead.

③ 5 objects $\rightarrow$ 5 connections.

④ Higher page-load time

Persistent

① Connection can be reused.

② Less overhead.

③ Can use one connection

④ Lower page-load time.

(B) Given, $RTT = 100 \text{ ms} = 0.1 \text{ s}$

There are 5 objects

So $2RTT$ per object (Non-Persistent)

$$D_{np} = 5 (2RTT) = 10 (100 \text{ ms})$$

$$D_{np} = 1000 \text{ ms} = 1 \text{ s}$$

$$D_p = 1RTT + SRTT$$

$$= 6RTT$$

$$= 6(100 \text{ ms})$$

$$D_p = 600 \text{ ms} = 0.6 \text{ s}$$

Ans. 13 An email message consists of header lines, a blank line, and the message body.

① Header lines: The headers contain information about the email, rather than the main content.

Common header includes: From, To, Date, Subject.

② Blank line: A blank line separates the header section from the message body.

It indicates that the header section has ended.

③ Message Body: The body contains the actual content of the email, such as text or other message content.